FYP Ideas

1. GridWatch

- Electricity theft (illegal meter tampering, direct line hooking) and unplanned load shedding cause major revenue losses and unstable supply for power distribution companies.
- Existing detection relies on manual inspections and complaints, which are slow, inconsistent, and catch only a small fraction of actual theft cases.
- System analyzes consumption patterns (using smart meter or grid data) to flag anomalies that indicate likely theft or tampering, such as sudden unexplained drops in billed usage.
- Same data can help predict and manage load shedding more fairly, identifying which areas are under real supply stress versus which are simply drawing power illegally.
- Gives utility companies an early, data-driven way to catch losses and plan outages more intelligently, instead of reactive, complaint-based responses.

2. CodeSense

- As codebases grow with multiple developers, quality silently degrades (long methods, deep nesting, duplicated logic) since no one sees the big picture.
- Manual code reviews are slow and inconsistent; existing linters only catch surface-level syntax issues, and design flaws often slip through during pull requests.
- A GitHub-integrated bot automatically reviews code on every push and pull request, posting maintainability and design-quality reports directly as inline PR comments, no manual uploads required.
- Combines static analysis metrics with LLM-based reasoning that understands intent (e.g., inefficient data structure choices, poor design patterns), triggered via GitHub webhooks.
- The bot explains why something is a problem and suggests a practical fix with diff-ready code snippets, giving teams a consistent quality baseline that enforces standards before merge.

3. Wall Moisture Early Warning System

- Wall dampness from leaks or poor drainage stays hidden until visible damage (peeling paint, mold) appears, by then the structure is already weakened.
- No easy, low-cost way exists for households to monitor in-wall moisture before damage becomes visible.
- Sensors placed at moisture-prone spots (bathrooms, kitchens, exterior/basement walls) continuously track dampness.
- App shows simple visual status per wall/zone (green/yellow/red), with caution and critical alerts.
- Same early-warning philosophy as the electricity idea, applied to structural health instead of billing.

4. FairPrice PK

- Real estate pricing in Pakistan is opaque, buyers/sellers rely on asking prices and informal comparisons, not data.
- Sites like Zameen.com host huge listing data but have no feature to answer, 'what should this house actually be worth?'
- Scrapes property listings (price, location, size, rooms, amenities) and trains an ML model on the relationship between features and price.
- User enters property details and gets a predicted fair market price range with comparable listings.
- Useful for buyers, sellers, and investors making data-driven pricing decisions instead of guesswork.

5. Agentic Sites Support Role

- Construction projects have many manual, high-pain processes (Document Control, Procurement, Site Planning, Contracts & Risk, etc.) that are candidates for AI-driven efficiency.
- A prior phase already identified the highest-priority workflows worth automating.
- Role helps move from 'identified opportunity' to working MVP, tracking AI pilot development.
- Helps teams actually adopt tools like Microsoft 365 Copilot and build simple automations (Power Automate, Copilot Studio).
- Also drives broader AI adoption across teams through training and support on Copilot, Claude, and other LLM tools.

6. Startup Mentor

- Aspiring entrepreneurs, especially in Pakistan, lack access to real business mentorship; generic online advice isn't locally relevant.
- Conversational AI mentor trained on startup knowledge, business strategy, and Pakistan-specific market insights.
- Combines NLP, a curated knowledge base, and an LLM for practical, context-aware guidance.
- Designed to be low-cost and feasible to build using free/low-cost APIs.
- Positioned as a socially useful project supporting local startups and entrepreneurs.

7. AwazComplaint

- ❖Many citizens, especially in rural or low-literacy areas, struggle to file complaints (utility, municipal, consumer) through text-based forms or apps.
- ❖Existing complaint portals assume comfort with typing, navigating menus, and written language — excluding a large portion of the population.
- ❖System lets users file a complaint simply by speaking in their local language, with speech-to-text and NLP converting it into a structured complaint entry.
- ❖Automatically categorizes the complaint (e.g. by department, urgency, location) and routes it to the right authority without the user needing to know where to send it.
- ❖Makes the complaint process accessible to illiterate/semi-literate users, increasing civic participation and giving authorities better-structured data on recurring issues.

8. NAVICODE

- ❖Developers joining a new codebase often face thousands of files with no clear picture of how they connect, spending weeks just reading before making a single fix.
- ❖Existing tools (GitHub, IDEs) let you browse code but don't show the big-picture structure or how a change in one file will ripple into others.
- ❖Core idea: paste a GitHub link (or upload a ZIP) and the tool reads the entire codebase to generate an interactive visual map of files and their connections.
- ❖Users can search in plain English (e.g. "where do we charge credit cards?") and the map highlights the exact relevant file, with warnings when a file has many dependents.
- ❖Turns an unreadable, sprawling codebase into something a new developer can navigate and understand in minutes instead of weeks.

9. SubSense

- ❖People accumulate recurring subscriptions across many emails/cards and usually only notice a forgotten one after months of wasted payments.
- ❖Existing expense apps show what you spent, not whether you're actually using what you pay for.
- ❖Scans email/bank statements (or manual CSV) to detect recurring charges automatically.
- ❖Cross-references each subscription against actual usage signals (app last-opened data, manual check-ins).
- ❖Surfaces a 'cost-per-use' score instead of a plain spending total, flagging unused subscriptions clearly.

10. Data Talk

- ❖Most real-world data goes unused due to two barriers: it's messy, and even when clean, extracting insights requires SQL knowledge.
- ❖Existing tools solve only one side like spreadsheets need manual cleaning, BI tools assume clean data, enterprise Text-to-SQL is costly.
- ❖User uploads raw data; the system automatically cleans it, then answers plain-English questions with visualized results.
- ❖Cleaning stage handles type correction, outlier detection, and smart missing-value handling without deleting data.
- ❖Querying stage converts natural language into safe, schema-aware SQL to prevent hallucinated columns, returning charts or tables.